

Adam Lin

Toronto, ON, Canada | Software Engineer | Open to remote roles in the Greater Toronto Area

PROFESSIONAL SUMMARY

Software Engineer with experience building distributed systems, full-stack web applications, and performance-critical backend services. Currently a Software Development Engineer at AWS working on Amazon RDS Aurora control-plane systems, with hands-on experience improving scalability, reliability, efficiency, and observability in large-scale cloud infrastructure. Strong background in Java, Rust, JavaScript/TypeScript, React, Node.js, Linux systems, SQL-backed services, and production operations.

TECHNICAL SKILLS

Languages:	Java, Rust, JavaScript, TypeScript, Python, C++
Backend / Platform:	Node.js, Spring, Quarkus, GraalVM Native Image, API design, service-to-service communication, streaming systems, asynchronous systems, distributed systems, microservice-style architectures
Frontend / Product:	React, Next.js, Angular, Tailwind CSS, responsive UI, web application development, user-facing product features
Cloud / Infrastructure:	AWS, Amazon Linux 2, Linux systems, observability, monitoring, alerting, incident response, on-call operations, infrastructure reliability, performance tuning
Data / Persistence:	Amazon RDS, SQL, MySQL and PostgreSQL (from Aurora domain experience), MongoDB, DAO design, data access patterns
AI-Assisted Development Tools:	GitHub Copilot, ChatGPT, Kiro, Amazon Q, Codex
Additional Relevant Exposure:	React Native (project familiarity), Kotlin (willing to ramp quickly), gRPC/protobuf concepts, Kubernetes fundamentals, Terraform familiarity, AI/agent-oriented development exploration

PROFESSIONAL EXPERIENCE

Amazon Web Services (AWS) - Software Development Engineer

Nov 2024 - Present | Toronto, ON

- Engineer on the Amazon RDS Aurora control plane, developing and operating distributed services that manage large-scale production database infrastructure
- Build and maintain always-on backend systems with a strong focus on scalability, stability, reliability, and efficient resource utilization
- Worked extensively in Java on backend services, contributing to feature development, service optimization, and reliability improvements in a large production codebase
- Key contributor to a major Aurora Serverless v2 initiative enabling scaling down to 0 ACU by developing minimal Rust agents to replace heavy Java components in the Aurora control plane
- Key contributor to replacing polling-based communication with streaming updates across both Java and Rust components, reducing CPU utilization by 10% and improving update speed by up to 15x
- Implemented authorization for the streaming update system, including message signing, validation, and encryption, ensuring secure service-to-service communication across distributed components
- Built a lightweight Rust replacement for a resource-heavy Java monitoring agent, reducing overhead while improving responsiveness, operational simplicity, and efficiency
- Core contributor to migrating a major service from Spring/Tomcat on the JVM to Quarkus with GraalVM Native Image, achieving 2x reduction in RSS memory footprint and approximately 20x faster startup times
- Developed strong experience with performance profiling, Linux production debugging, runtime behavior, memory usage, startup optimization, and production tradeoff analysis
- Built practical experience with AWS IAM and secure access patterns while developing and operating control-plane services, including work involving roles, grants, and trust policies
- Participated in 24/7 on-call rotation, leading incident mitigation, debugging live production issues, and authoring root cause analyses
- Contributed to design reviews and code reviews, providing feedback on maintainability, correctness, safe rollout practices, and long-term operational health

Lume Community Inc. - Volunteer Full Stack Engineer

Apr 2023 - Present | Remote

- Built and shipped user-facing product features using React, TypeScript, and Next.js
- Helped develop a responsive sign-up web experience and contributed to end-to-end feature delivery in a collaborative, small-team environment
- Worked closely with teammates across product thinking, implementation, and technical planning
- Maintained responsive, accessible UI with Tailwind CSS and contributed to customer-facing web application quality
- Supported engineering workflow and delivery processes through GitHub Projects, issues, milestones, and team coordination

Level Access - Full Stack Engineer Intern

May 2022 - Aug 2022 | Toronto, ON

- Developed full-stack features using the MEAN stack, working in JavaScript across MongoDB, Express, Angular, and Node.js
- Built frontend functionality in Angular and implemented backend validation for critical workflows and user-facing application flows
- Contributed to a product focused on accessibility and compliance, helping improve the usability and quality of web experiences
- Designed and implemented Data Access Objects (DAOs) for MongoDB collections
- Wrote unit and integration tests using Jest and Karma
- Participated in code reviews and Agile ceremonies while collaborating in a professional engineering team

Nick's Method - Math & Computer Science Tutor

May 2022 - Nov 2024 | Toronto, ON

- Taught Java and Python, emphasizing programming fundamentals, debugging, clean code, object-oriented design, and software engineering best practices
- Helped students work through unfamiliar technical problems, improve code quality, and complete multi-step projects

Alpha Coding - Computer Science Instructor

Aug 2020 - Mar 2022 | Ontario, Canada

- Taught programming in Python, Java, and Scratch to elementary and high school students
- Designed lesson plans, exercises, and project-based learning materials

EDUCATION

Carleton University - Bachelor of Science in Computer Science

Jan 2019 - May 2024

Scholarships: Carleton Entrance Scholarship; Computer Science Excellence Scholarship

LANGUAGES

English (Native) | French (Limited Working Proficiency)